

Demonstration of Advanced Mars ISRU CO₂ Recovery System

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Human exploration of Mars and unmanned sample return missions can benefit greatly from the resources available on Mars. The first major step of any Mars *in-situ* propellant production system is the acquisition of carbon dioxide and its compression from a total pressure of around 5 torr to a reasonable storage and processing pressure in the 100 to 200 kPa range. This compression ratio of ~150 (from 5.0 to 760 torr) can be achieved by common vacuum pump technology, however, operating such devices in the Mars environment is difficult and unreliable due to the limited lifetime of rapidly moving parts, temperature extremes, frequent starts and stops, and dust. The power requirement, in the form of electricity, is also prohibitive unless the spacecraft has nuclear power. TDA Research Inc. is developing a compact, lightweight, advanced sorbent-based compressor to recover high-pressure, high purity CO₂ from the Martian atmosphere. The system eliminates the need for a mechanical pump, increasing the reliability with relatively low power consumption. TDA's system uses a new, high capacity sorbent that selectively adsorbs CO₂ at 5 torr and regenerates by temperature swing, producing a continuous, high purity CO₂ flow at pressure (> 760 torr). Previously, we have successfully completed bench-scale proof-of-concept demonstrations, elevating the TRL to 3. In a current SBIR Phase II project, we further scaled-up the sorbent production and are now working on integrating the sorbent into a sub-scale rapid (relatively fast, about 15-20 min) thermal swing adsorption/desorption cycling prototype system.

Nomenclature

Ar	=	Argon
CFD	=	Computational Fluid Dynamics
CO ₂	=	Carbon dioxide
gpm	=	gallons per minute
H ₂ O	=	Water
hr	=	hour
I.D.	=	Inner Diameter
ISRU	=	in-situ resource utilization
L/D	=	ratio of Length/Inner Diameter of the reactor
MOF	=	metal organic framework
N ₂	=	Nitrogen
O ₂	=	Oxygen
P _{des}	=	desorption/recovery pressure
PLSS	=	Portable Life Support System
SBIR	=	Small Business Innovation Research
T _{ads}	=	adsorption temperature
T _{des}	=	desorption temperature
wt.	=	weight

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I. Introduction

TO greatly reduce the amount of resources brought from earth to Mars, in-situ resource utilization (ISRU) on Mars will use materials locally available to produce valuable fuels and other consumables. Carbon dioxide (CO_2), present at low pressure (~ 5 torr) in the Martian atmosphere, is one such local material that can provide carbon and oxygen to be reacted with hydrogen brought from earth. To make use of the Martian CO_2 , the gas needs to be collected from the atmosphere at low pressure and compressed to be used at a higher pressure (200 kPa or more). Conventional pumps can readily achieve this, but may not be suited to the Mars environment, both in terms of maintenance and power requirements.

TDA is developing a compact, energy efficient adsorbent-based solid-state compressor that uses a metal organic framework (MOF) to collect and concentrate dilute CO_2 from the Martian atmosphere. This sorbent is highly selective to CO_2 and is very stable in the presence of other constituents present in the Mars atmosphere (argon, ozone etc.). The sorbent adsorbs CO_2 via strong physical adsorption (the heat of adsorption is 11.5 kcal/mol).¹ This relatively strong attraction enables the sorbent to achieve high adsorption capacity even when CO_2 is present at very low concentrations, but because no covalent bond is formed the CO_2 can be removed and the sorbent can be easily regenerated by changing the operating conditions (i.e., applying heat). We have so far demonstrated stable performance over 3,500 thermal swing adsorption/desorption cycles. We measured a CO_2 delivery of 1.3 mol/kg sorbent at 70 psia under temperature swing cycling of the bed temperature from -78°C to 100°C . We scaled-up the sorbent production and designed a sub-scale rapid (relatively fast, 15-20 minute) thermal swing adsorption/desorption cycling prototype system. The prototype system fabrication is being carried out and results from the prototype system demonstration will be provided at the next ICES meeting.

II. Adsorption Isotherms and CO_2 Selectivity

We measured the CO_2 , N_2 and Ar adsorption isotherms on the MOF sorbent in the high resolution isotherm measurement unit (Micromeritics ASAP 2020). This unit is equipped with multiple pressure transducers that allow us to precisely measure the CO_2 , N_2 and Ar adsorption data at low partial pressures of interest. Figure 1 shows the CO_2 , N_2 and Ar adsorption isotherms on MOF adsorbent at 0 and 30°C . We see that the MOF sorbent has excellent selectivity ($> 1,000$) for CO_2 over both N_2 and Ar at 5 torr.

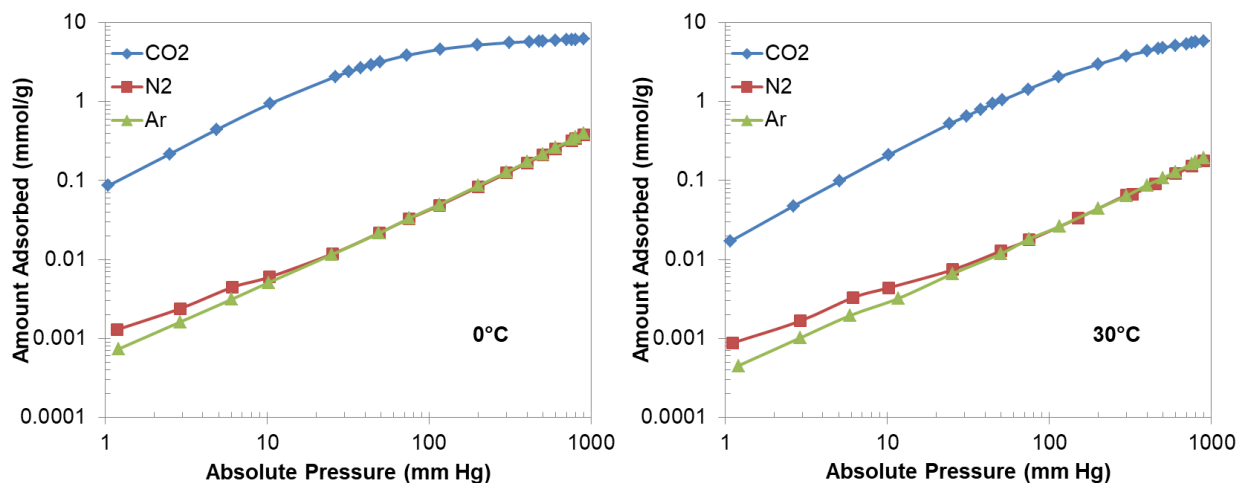


Figure 1. CO_2 , N_2 and Ar adsorption isotherms on the MOF adsorbent at 0 and 30°C .

III. Fixed Bed Test Results

The test stand used has the capability to accurately measure the pressure rise produced during the CO_2 desorption at high temperatures. The test apparatus utilizes a reactor with a 0.62" inner diameter and a 2.1" length. The vessel is mounted within an insulated jacket that is cooled and heated using nitrogen as heat transfer medium. When sub-

ambient temperatures are desired, liquid nitrogen is directed into the jacket using cryogenic solenoid valves. To heat it nitrogen gas is passed through a heater cartridge to carry energy into the jacket of the reactor and heat the sorbent material within. The apparatus uses two mass flow controllers to control gas delivery into the reactor and a vacuum pump to reduce the pressure in the system to the desired test conditions.

During a typical cycle, the adsorption gases are directed through the sorbent bed at sub ambient pressure and temperatures until the material is saturated with the CO₂ and the outlet CO₂ concentration reaches 90%. During regeneration the sorbent vessel is initially sealed and the outer jacket is heated. The energy from heating dislodges this adsorbed gas from the adsorbent and begins to pressurize the sorbent vessel. The vessel is pressurized up to the delivery pressure set point and then additional gas that is desorbed is metered out of the bed and vented through a third mass flow controller. Figure 2 shows the bed temperature, pressure and the CO₂ concentration in gases exiting the reactor during a typical cycle adsorption/ desorption cycle.

The outlet gas flow is then integrated so that the total moles of the pressurized gas leaving the system can be determined, which is then divided by the mass of adsorbent loaded into the system to determine the loading. The purity of the CO₂ outlet after regeneration was quantified using a Quantek Instruments CO₂ detector. This system is capable of rapid thermal swing cycles and our tests have a full cycle time of 40-70 minutes when cycling between -78°C and 100°C and as low as 17 min in a -20 to 50°C cycle.

A. Testing with Reactors of Different Diameter

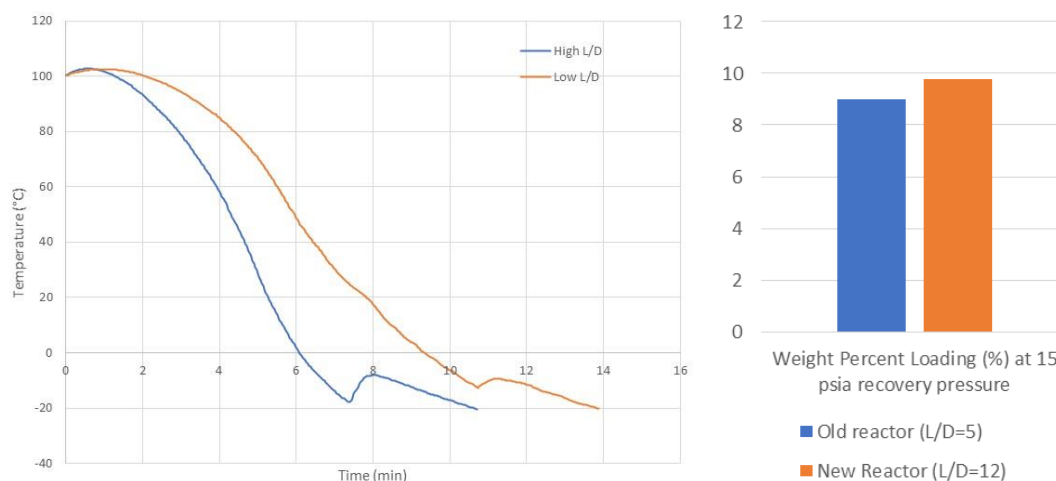


Figure 3. Impact of reactor dimensions. The figure on the left compares the cooling rates of the two reactors and the graph on the right compares the CO₂ production from the two beds.

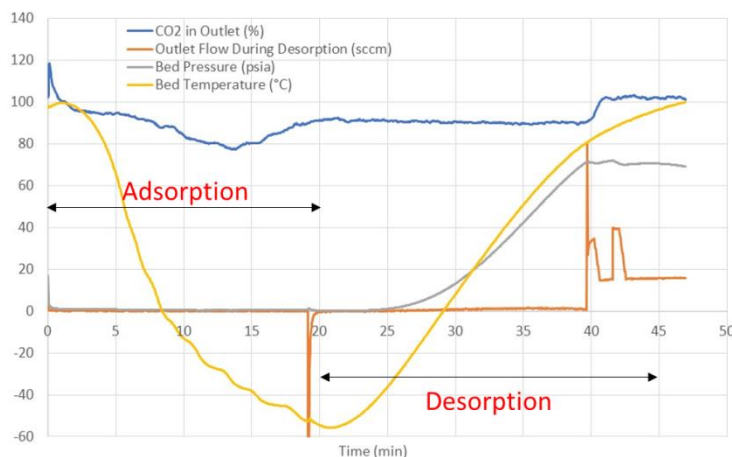


Figure 2. A typical adsorption and desorption cycle. The gas flow during adsorption step (time 0 to ~20min) is 5%Ar/95%CO₂ at 2-7 torr. The concentration drop during this period shows the CO₂ uptake of the material. Next, during the desorption step (time 20 to ~40 min) the bed is heated and plugged. As the temperature rises, the pressure rises as well until the desired pressure (~70 psia) is reached in the bed. The outlet flow controller then opens and the additional gas that desorbs flows out while the flow controller varies this outlet flow to hold the system pressure above (in this test) 70psia as long as possible.

To better understand the effect of reactor dimensions on both the cooling rates (heat transfer) of the material as well as the performance (mass transfer), two reactors were tested with the same sorbent sample. The initial reactor had an internal diameter and bed depth such that the L/D (length divided by diameter) was 5 (0.67" I.D.) while the new reactor had a L/D of 12 (0.46" I.D.). The expected advantages of the higher L/D (and a lower I.D.) are quicker cooling and heating rates for the bed from lower heat transfer distance and better CO₂ removal efficiency of the bed during adsorption from higher bed depth. The faster cooling and heating will translate to much faster cycles, which is desirable. However, this is limited because as the L/D increases, the pressure drop through the bed in the final design will increase as well.

Figure 3 compares the cooling rate of the sorbent bed for the two reactors. The jump in the temperature towards the end of the cooling cycle in both curves occurs when the adsorption gas and vacuum are turned on and comes as the heat generated by adsorption is pushed out of the bed by the gas flow. As can be seen in this figure, the reactor with the higher L/D (smaller bed diameter) has a cycle time that is about a third shorter than the larger diameter reactor. The graph on the right shows the amount of the CO₂ released (i.e., CO₂ working capacity of the sorbent on a % wt. basis) and the bed with the larger L/D showed higher capacity.

B. Impact of Decreased Time of Flow

During cool down the beds do not adsorb CO₂ till they get down in temperature below a certain limit, hence we varied the temperature at which the adsorption gas flows start to maximize the sorbent loading. Also, we would like to minimize the power consumption of the CO₂ harvesting system by reducing the amount of gas that is moved through the system. The test was altered such that during the initial cooling the bed was exposed only to the CO₂ remaining in the bed from the previous regeneration. Once the bed had cooled to a specified temperature the vacuum and the gas flow, mimicking the Mars atmosphere, were initiated and the bed continued to cool. Once the bed reached the desired minimum temperature, the bed switched to the typical heated regeneration. The goal was to increase the amount of CO₂ adsorbed relative to the amount of CO₂ introduced to the bed during the adsorption.

Figure 4 shows the results of this test. The top graph shows the percent of CO₂ recovered at pressure with respect to the amount added to the system during adsorption at the different temperatures the gas and vacuum were applied. As the temperature at which the gases and vacuum were turned on decreased (corresponding to a shorter time of flow) the recovery increased, as expected (the CO₂ adsorption wave fronts have not fully progressed (i.e., not broken through), resulting in higher recovery). It is important to note (that the higher recovery rated does not necessarily correspond to higher loadings as shown in the bottom graph in Figure 4; this is due to the shorter time of flow. At the lowest gas and vacuum on temperature tested, the loading showed a significant decrease. However, this is an artifact of the way the cycles are

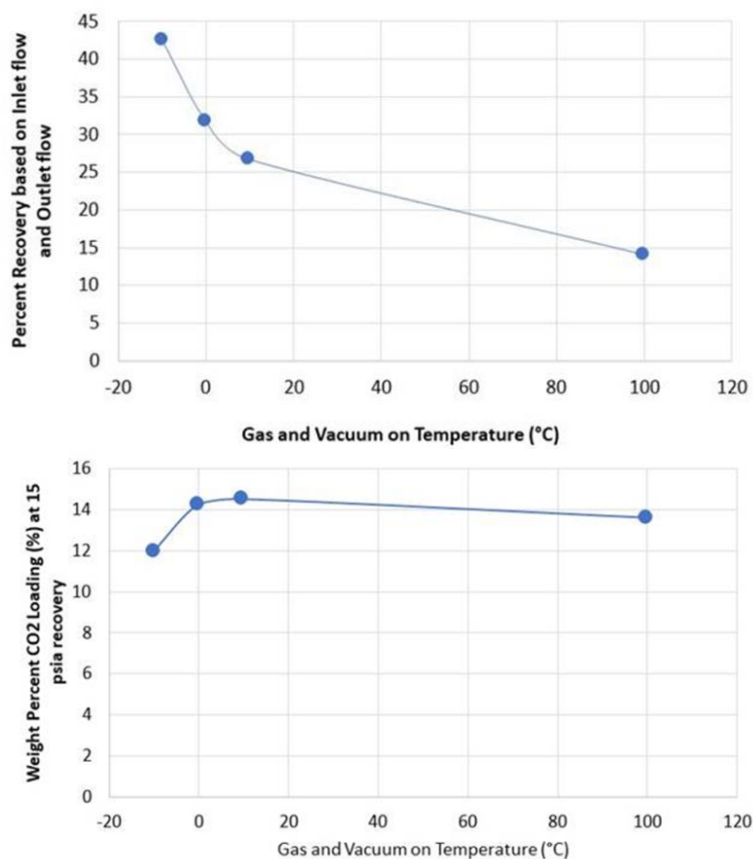


Figure 4. Comparison of the recovery rates for tests with truncated gas flow during adsorption. During cooling the inlet gas and vacuum were started at different temperatures. Adsorption temperature: $T_{ads} = -20^{\circ}\text{C}$, Desorption Temperature: $T_{des} = 100^{\circ}\text{C}$, Desorption/Recovery Pressure: $P_{des} = 15$ psia.

controlled in our bench scale tests, for e.g., if the gas flow was continued longer for this condition the capacity would have increased and the sorbent probably would achieve the same working capacity.

C. Impact of Space Velocity

Next, we varied the space velocity of the gas flow during the adsorption step and the results are presented in Figure 5. In this test the sorbent bed was cycled between -20°C and 100°C , with the gases and vacuum starting at -10°C during the cooling cycle and the desorbing CO_2 was recovered at 15psia.

We compared the CO_2 working capacity and the percent CO_2 recovered at pressure. A higher working capacity indicates a more effective use of the sorbent (smaller beds), but a higher recovery indicates that the system is used more efficiently (less power draw). We observed that as the flow increases, the amount of CO_2 adsorbed increases as well, but the efficiency of the system, as measured by the recovery rate decreased. This indicates that there is a trade-off between working capacity and recovery and we have to operate this thermal swing based solid state compressor system at an optimum condition.

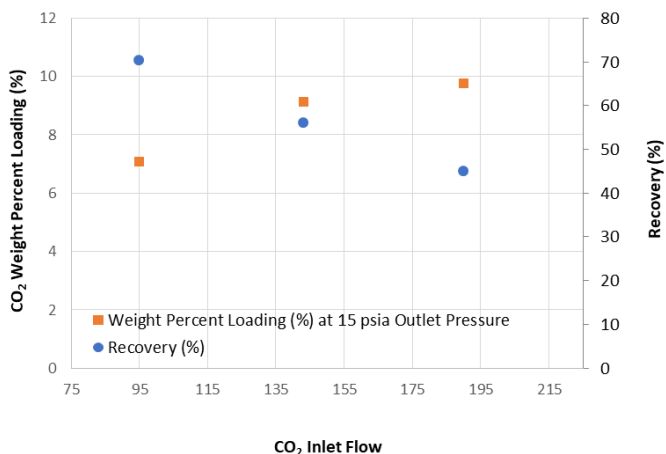


Figure 5. Comparison of the recovery rates for tests with truncated gas flow during adsorption. During cooling the inlet gas and vacuum were started at different temperatures. Adsorption temperature: $T_{\text{ads}} = -20^{\circ}\text{C}$, Desorption Temperature: $T_{\text{des}} = 100^{\circ}\text{C}$, Desorption/Recovery Pressure: $P_{\text{des}} = 15 \text{ psia}$.

D. Life Test

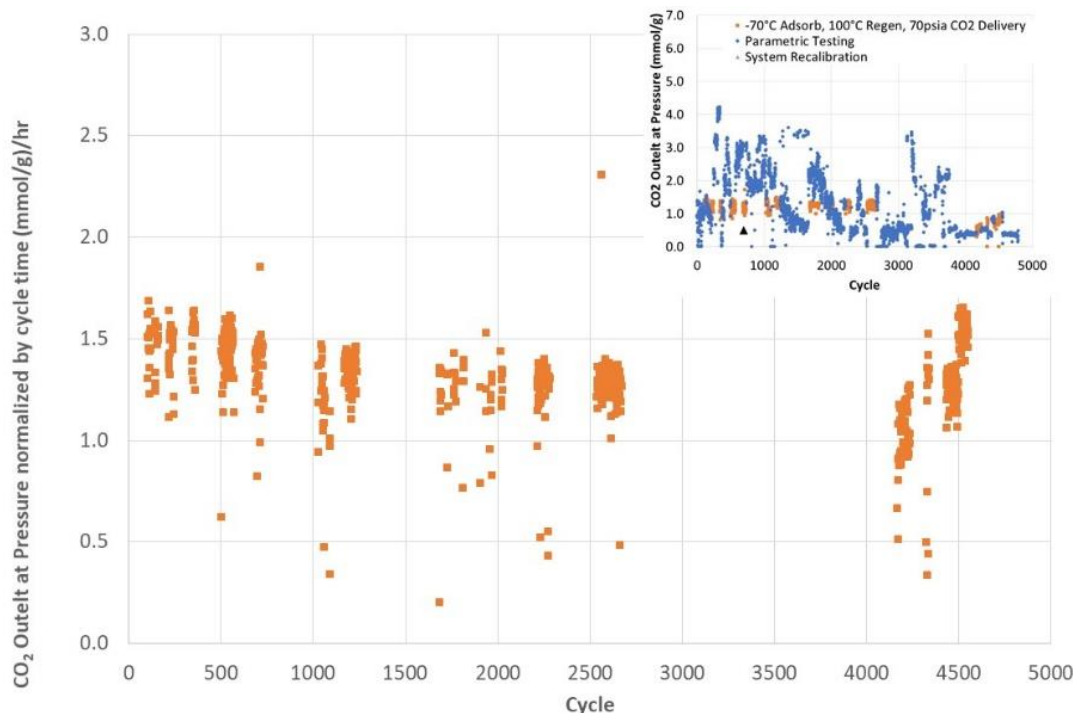


Figure 6. Cycles testing on the sample. Inset graph shows loadings for all the tests completed. The main figure shows tests done at a -70°C adsorption, 100°C desorption with a recovery pressure of 70 psia. These results are normalized by cycle time to compensate for the reactor change.

Over the course of the parametric tests described in the previous sections, we ran the sorbent bed through 4,500 thermal cycles. The results from these cycles are presented in Figure 6. The majority of this data was taken at different conditions as part of the parametric tests, so most of the points are not useful for measuring the stability of the sorbent. However, we returned to a baseline condition several times over the course of the testing to assure that the sorbent material has retained its initial performance. The data shown in the main graph in Figure 6 (represented by the orange squares) was taken at this baseline condition ($T_{ads} = -70^{\circ}\text{C}$, $T_{des} = 100^{\circ}\text{C}$ with $P_{des} = 70$ psia) while the entire 4,500 cycles are shown in the Figure 6 inset. As seen in these results the sorbent maintained a stable performance over the 4,500 thermal swing cycles achieving 1.5 mmol/g/hr working capacity under the baseline conditions illustrating that the sorbent material is very stable.

IV. Prototype Unit Design

For human Mars surface missions to return back to earth it has been estimated that approximately 30 metric tons of oxygen and methane propellant are required.² To make this much propellant approximately 20 metric tons of CO_2 and 16 metric tons of water (H_2O) is required from ISRU systems. Current ISRU guidelines call for three independent modules, each capable of producing 50 percent of the total propellant needed, thereby providing capability to recover from a complete failure of one module.³ Based on module redundancy, mission assumptions, and ascent propellant amounts needed, per module production rates are 1.1 kg O_2 /hr for O_2 produced only from the Martian atmosphere, which corresponds to a CO_2 capture rate of 1.51 kg CO_2 /hr.

Using the above mentioned basis and the capacity achieved in our baseline test conditions, we estimated the amount sorbent required for a single module as a function of cycle time. We then sized the sorbent beds and estimated the system weight excluding the valves and the heat exchange equipment. We also calculated the amount of heating/cooling that needs to be supplied to the system and then estimated the specific heat transfer rate (W/m^2), if we use simple fixed beds heated and cooled by a jacket. We observed that the optimal cycle time in terms of the system weight (excluding heat transfer components) is between 30 min to 120 min (0.5 h to 2 h). The heating/cooling needs for the sorbent system are not dependent on cycle time (about 0.91 kW) with only small changes coming from difference in heat losses between different size systems. The ratio of heat transfer equipment weight (auxiliary units needed to carry out thermal swing cycles) to the sorbent system weight will be much higher for the smaller system compared to larger system, since the heat duty (heat transfer needs are similar) is similar. However the smaller system with active heating/cooling will reduce the overall system weight and have higher reliability than a much larger passive system that is sized to coordinate with Martian diurnal cycles. Hence, we sized our sub-scale unit to be $1/10^{\text{th}}$ of the full scale module, i.e., capable of producing 0.15 kg CO_2 /hr.

V. Reactor Design Optimization

The thermal conductivity of the MOF sorbent material is not as high as that of carbon based materials, so we explored multiple reactor designs with the main goal of maximizing the heat transfer while still offering low pressure drop through the sorbent bed. We started with a reactor layout similar to one we previously built for a NASA SBIR project (CO_2 Removal from Mars EMU, Contract: NNX11CB43C). The basis of this reactor revolved around using the astronaut's metabolic heat as source of energy to thermally swing the PLSS CO_2 sorbent bed. Implementing what we learned from that project, a revised version of this reactor was developed. Figure 8 shows a preliminary design for the reactors that can be 3-D printed in aluminum. This allows us to build a near completely

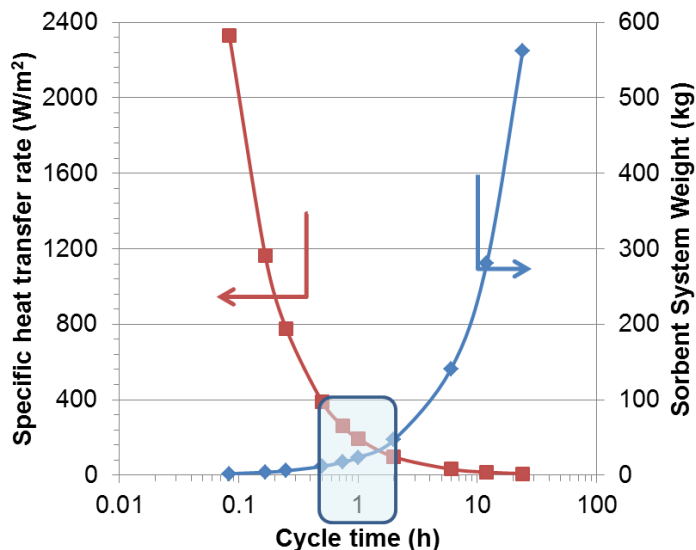


Figure 7. System size and heat transfer needs under rapid thermal swing cycles.

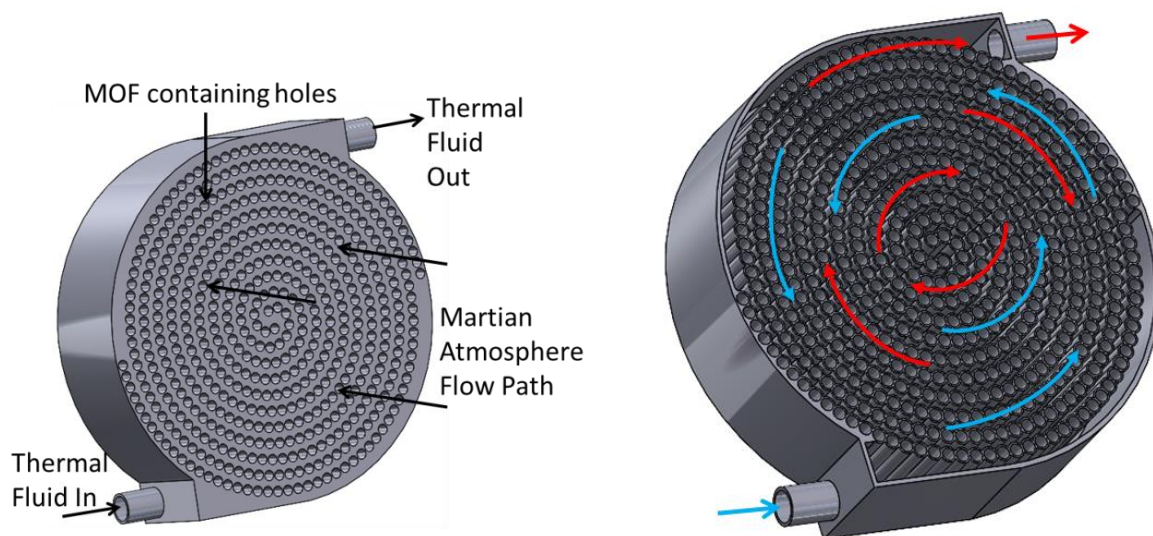


Figure 8. Revised maximum heat transfer sorbent reactor (left), cut a way showing cooling water path (right).

hollow reactor (Figure 8 right) that is similar to a shell and tube heat exchanger. The tubes are for the sorbent and the shell side is for the heat transfer liquid. This design allows us to mimic the small bench scale sorbent beds that were tested and design the prototype unit as just 100s of modules that operate in parallel. Since the sorbent will reside in small (0.25-0.5" diameter) holes it vastly decreases the mean free path for heat to enter and leave the sorbent while also increasing the surface area between the highly conductive aluminum walls and the poorly conductive sorbent material. Thermal fluid will flow around the holes in a spiral pattern around one side of the holes until it gets to the center of the reactor which then will flow on the opposite side of the holes and out the reactor. This fluid flow design allows for the greatest contact between the sorbent walls and the fluid, thus significantly reducing the resistance to thermal flow between the fluid and reactor walls. This reactor geometry would be nearly impossible to fabricate using traditional manufacturing methods. However, with the advances in 3-D printing technology these designs can now become a reality.

To further develop the design, we carried out CFD modeling to determine if the proposed heat transfer scheme will work effectively. We used Solidworks 2019's Flow Simulation package to simulate flowing 2 gpm of 3M's NOVEC 7000 fluid through the reactor. We solved the simulation on a time dependent basis where at time 0 sec, the reactor and fluid inside is at 100°C (bed has completed desorption) and -40°C fluid then begins to flow. Figure 9 shows that the reactor walls can be effectively cooled through a temperature swing of 140 degrees in as little as 70

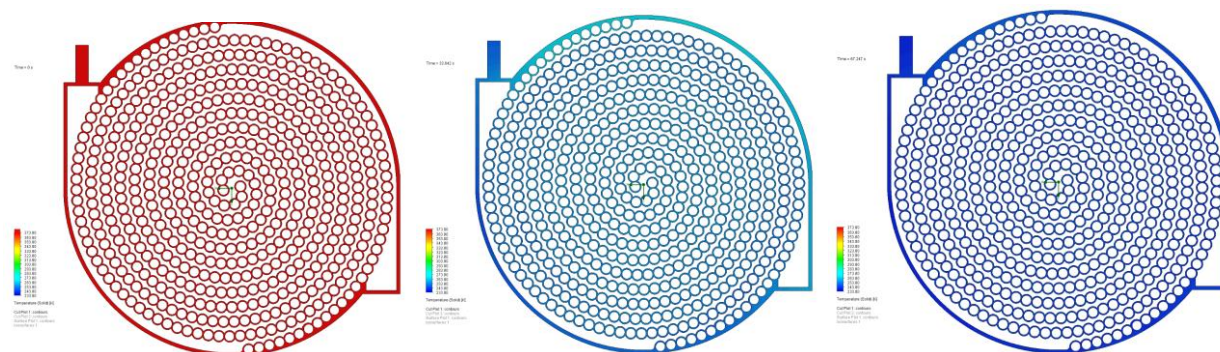


Figure 9. Time dependent CFD reactor modeling T=0 sec (left), T=30 sec (middle), T=70 sec (right).

seconds. This initial simulation did not account for the sorbent media being present inside the tubes/holes.

Next, we worked include the sorbent media in our CFD models. One if not the most important aspects in the heat transfer modeling and design is the resistance in heat transfer between the reactor walls and the MOF sorbent

material. This has been shown to make up the bulk of the resistance in the overall heat transfer coefficient in the reactor. In order to accurately model this heat transfer, we used temperature data from the bench scale reactor tests. We internally installed a 1/16" thermocouple in our bench scale reactor to measure the bed temperature of the MOF. We attached a wire thermocouple to measure the reactor surface (wall) temperature (Figure 10). We then made a representative 3D model in Solidworks of the 316 stainless steel reactor tube and sorbent material. We set up a simple 2D computational domain encompassing the tube and MOF sorbent and in an effort to speed up computational time; we assumed symmetry in the Z direction. For a boundary condition, we fixed the reactor wall temperature profile to our experimental data. To model the MOF material, we created a custom porous media material in Solidworks Flow Simulation. We used measured values for the density, porosity and heat capacity. We then altered the thermal conductivity of the material until the average temperature of the MOF in the CFD model closely fit the bed temperature profile of the actual MOF material from bench scale tests. The comparison of temperature profiles can be seen in Figure 11. We able to fit the experimental with our CFD model within a 5°C degrees difference between the predicted and the observed temperature for the MOF sorbent bed.

Using the empirical fitting parameter (here thermal conductivity) from the CFD model obtained with the bench scale reactor tests, additional simulations were carried out on sorbent cells of varying diameters to see what effect this has on the rate of temperature change in the sorbent cell. In these simulations we kept the same wall thickness and sorbent volume. The flow rate of coolant going past the cell as well as its temperature was also kept the same in each simulation case. The only difference between them are the reactor lengths (and heat transfer area i.e., tube surface area contacting the heat transfer fluid) which were altered to achieve equal sorbent volume. The results in Figure 12 show the rate of cooling for sorbent in 1/2", 5/8", and 3/4" outside diameter cells. The surface temperature of the larger diameter cells cool down at a faster rate due to the lower mass per volume of the cell wall however, this does not translate to a faster cool down of the sorbent within the cell. This is to be expected since the MOF's thermal conductivity is so poor it takes longer for heat to be transferred through a larger diameter sorbent bed to the center of the cell. Hence, we concentrated on smaller diameter tubes for the shell and tube sub-scale prototype reactor modeling.

Figure 13 shows a time dependent Solidworks Flow Simulation for the cooling and heating of the spiral reactor containing 3/8" sorbent cell diameters. The initial conditions for the sorbent bed, metal walls and heat transfer and the gas inside the reactor all begin a temperature of 100°C (i.e., after desorption/regeneration step). At time 0 seconds, 2 gpm of NOVEC 7500 fluid

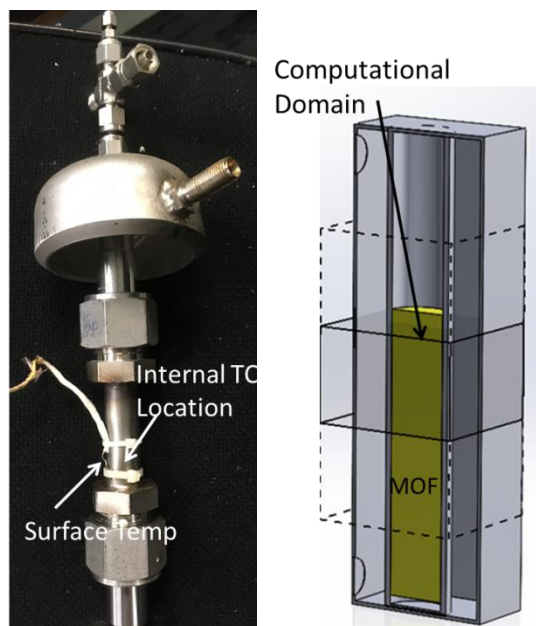


Figure 10. TDA's test reactor (left), Solidworks CFD model (right)

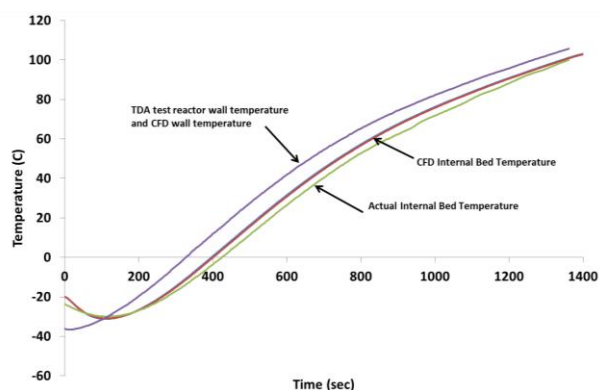


Figure 11. CFD results for MOF temperature results compared to actual bed temperature data.

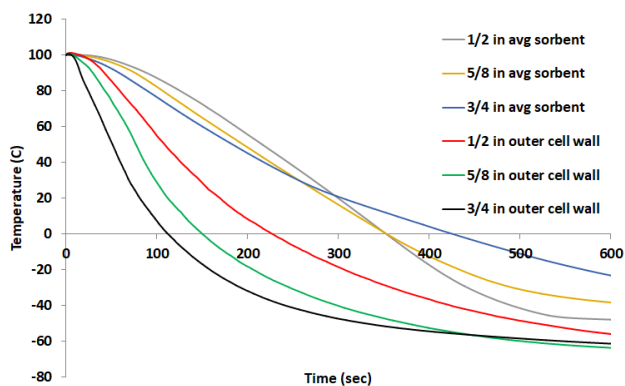


Figure 12. CFD results for MOF temperature at different diameter cells

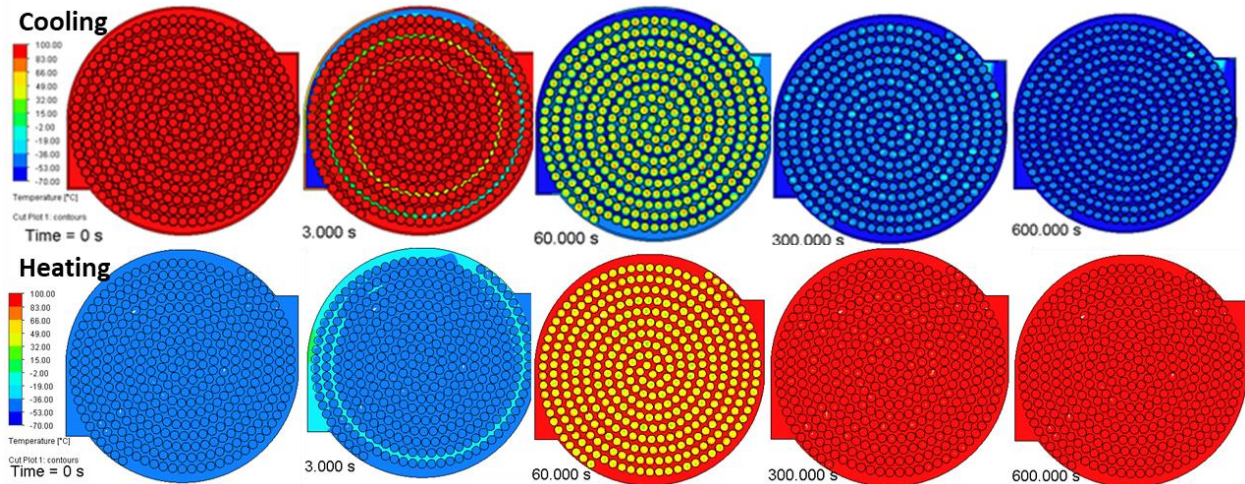


Figure 13. Time dependent cooling (top) and heating (bottom) of the spiral reactor design with 3/8" sorbent cells.

at -70°C begins flowing through the reactor. After 300 seconds, the entire reactor and sorbent is at -40°C and by 600 seconds the reactor reaches its asymptotic lowest temperature of -50°C. The simulation was then continued to simulate the switching to desorption i.e., 100°C NOVEC 7500 fluid at the same flow rate as the cooling fluid (2 gpm).

The result from one complete thermal cycle in the CFD simulations is shown in Figure 14. The temperatures of 3 sorbent cells were tracked during the simulation, one at the fluid inlet, in the middle of the reactor and one that is at the last of the spiral. The thermal profile show relatively consistent temperature profiles amongst the sorbent cells. Surprisingly, the inlet sorbent cell had the lowest rate of cooling during the simulation but was consistent with the other cells during reactor heating. When comparing the CFD velocity data of the fluid, this cell was subjected to the lowest velocity fluid during the cooling cycle. This was a result of the wide plenum for the inlet which causes the fluid to be more stagnant around these cells. This was especially apparent during the cooling since the viscosity of cryogenic fluid is much lower than it is during the heating. The reactor design is being further optimized to ensure that the velocity is more consistent throughout the entire reactor before we finalize the design for fabrication.

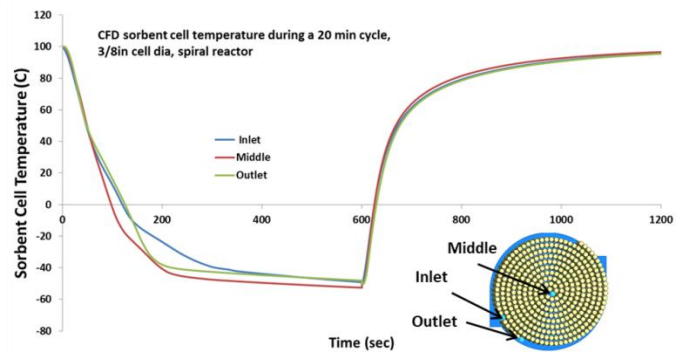


Figure 14. Complete cycle CFD reactor temperature profile.

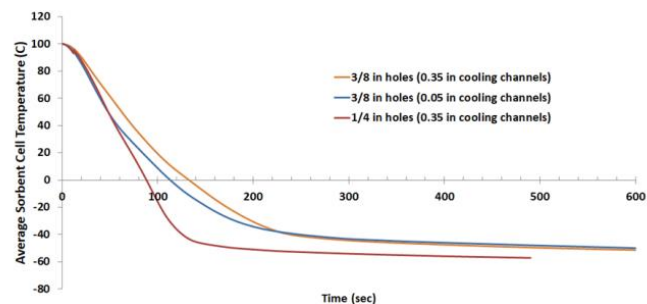


Figure 15. Effects of sorbent cell diameter on the sorbent cool down rate of the spiraled reactor

In addition to the 3/8" diameter sorbent cells, we also simulated the 1/4" diameter cells and compared their results for adsorption/cooling portion of the cycle. As shown in Figure 15, the reactor that contains 1/4 in diameter sorbent cells with large cooling channels (0.35 nominal spacing) cooled down the fastest. When the sorbent cell diameter was increased to 3/8" and leaving the cooling channel spacing the same as the 1/4" cell diameter it took approximately 100 seconds longer for the sorbent bed to reach -40°C. We then varied the width of the coolant flow channel, reducing it from 0.035" to 0.05" and kept the coolant flow rate to be the same. At the same coolant flow, the velocity increases and this provides more turbulence and reduces the boundary layer between the fluid and outer

sorbent cell walls. This configuration reduced the cooling time by 20 seconds but at the cost of increased power due to the larger pressure drop for coolant flow through the reactor as it takes approximately 70 psi more pressure to flow through the reactor.

VI. Conclusions

A mission to Mars will require fuel for planetary operations and return travel. Transporting sufficient fuel for these operations and the return journey is extremely costly and greatly increases the overall mass and complexity of the trip. The burden of transporting fuel to Mars can be greatly reduced by ISRU. Martian resources such as CO₂ and H₂O can be utilized on the planetary surface to generate fuels like methane or more complex hydrocarbons using processes like the Sabatier reaction. TDA has developed a novel material to adsorb the CO₂ from the Martian atmosphere and pressurize it for in-situ fuel production. We demonstrated excellent stability for the sorbent through 4,500 thermal swing cycles. We have also developed the reactor designs for the rapid thermal swing adsorption/desorption cycles and validated the design using CFD models. Next, we will finalize our reactor design and build the sub-scale prototype system to demonstrate rapid thermal swing cycle operation at a 1/10th scale of the full scale module.

VII. Acknowledgments

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VIII. References

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² Polsgrove, Tara, Thomas, Dan, Sutherlin, Steven, and Stephens, Walter, “Mars Ascent Vehicle Design for Human Exploration”, *AIAA Space 2015*, Pasadena, CA, 31 Aug-2 Sep, 2015

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